



Development and validation of the Dimensional Anhedonia Rating Scale (DARS) in a community sample and individuals with major depression



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ABSTRACT

Anhedonia, a core symptom of Major Depressive Disorder (MDD), is predictive of antidepressant non-response. In contrast to the definition of anhedonia as a “loss of pleasure”, neuropsychological studies provide evidence for multiple facets of hedonic function. The aim of the current study was to develop and validate the Dimensional Anhedonia Rating Scale (DARS), a dynamic scale that measures desire, motivation, effort and consummatory pleasure across hedonic domains. Following item selection procedures and reliability testing using data from community participants ($N=229$) (Study 1), the 17-item scale was validated in an online study with community participants ($N=150$) (Study 2). The DARS was also validated in unipolar or bipolar depressed patients ($n=52$) and controls ($n=50$) (Study 3). Principal components analysis of the 17-item DARS revealed a 4-component structure mapping onto the domains of anhedonia: hobbies, food/drink, social activities, and sensory experience. Reliability of the DARS subscales was high across studies (Cronbach's $\alpha=0.75$ – 0.92). The DARS also demonstrated good convergent and divergent validity. Hierarchical regression analysis revealed the DARS showed additional utility over the Snaith–Hamilton Pleasure Scale (SHAPS) in predicting reward function and distinguishing MDD subgroups. These studies provide support for the reliability and validity of the DARS.

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1. Introduction

In recent years, the assessment of anhedonia has seen a resurgence of interest. This has been borne out of findings from several large scale studies, including the Sequenced Treatment Alternatives to Relieve Depression (STAR*D) and the Genome Based Therapeutic Drugs for Depression (GENDEP), which have demonstrated that loss of interest (anhedonia) is a predictor of non-response to selective serotonin reuptake inhibitors (SSRIs) (Uher et al., 2008; 2012). This was also observed in an adolescent MDD sample, wherein among symptom dimensions, anhedonia

was found to be the only unique negative predictor of time to remission and depression free days with SSRI use (McMakin et al., 2012). Providing further generalizability across treatment modalities, anhedonia was a predictor of non-response to repetitive transcranial magnetic stimulation (rTMS) of the dorsomedial prefrontal cortex in MDD (Downar et al., 2014).

Notably, there is a paucity of measures and tasks validated for use in MDD that tap into the different facets of a “pleasure response”. Traditionally, anhedonia has been defined as a “loss of pleasure” (Ribot, 1986), a definition that emphasizes the consummatory/enjoyment aspect of reward function. However, anhedonia within the DSM-5 reflects a broader conceptualization that includes interest as well as consummatory pleasure. Even within the widely-used Hamilton Depression Rating Scale, a single anhedonia item is used to measure a dimensional construct

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including desire, effort and consummatory pleasure (Hamilton, 1960). This equivocal conceptualization makes anhedonia measurement difficult and imprecise. As Treadway and Zald (2011) assert “heterogeneity at the level of symptom definition is at least as problematic as...issues of comorbidity...” thus, refining this construct is imperative if we hope to understand the neurobiological underpinnings of anhedonia. Consequently, it may be more helpful to explicitly use anhedonia as an umbrella term for impairment of hedonic function over a spectrum of behaviors reflecting initial interest/desire (wanting a reward), anticipation (expectation of a reward), motivation (initial energy expenditure to attain a reward), effort (sustained energy expenditure to attain reward), and consummatory pleasure (enjoyment of reward).

The four main validated self-report measures used in clinical research to assess anhedonia are the Snaith–Hamilton Pleasure Scale (SHAPS) (Snaith et al., 1995), the Fawcett–Clark Pleasure Capacity Scale (FCPS) (Fawcett et al., 1983), the Revised Chapman Physical Anhedonia Scale (CPAS) and the Chapman Social Anhedonia Scale (CSAS) (Chapman et al., 1976). Ideally, a scale quantifying anhedonia in the context of MDD should be able to detect state versus trait differences, measure different aspects of anhedonia, distinguish between anhedonia and related constructs, and be appropriate for use in samples with a range of cultural beliefs and preferences (generalizability). These four measures differ in their ability to address these factors.

First, the CPAS/CSAS measure trait anhedonia (queries general response), whereas the FCPS and SHAPS measure state anhedonia (queries “right now” and “last few days”, respectively). It is still unclear whether anhedonia or its subcomponents are stable constructs over time in depressed patients (trait) or whether it is a symptom that fluctuates depending on severity or even antidepressant mechanism (state). While anhedonia is a symptom of a depressive episode, there is evidence to suggest that this symptom persists even in the remitted state (Nierenberg et al., 1999; Pedrelli et al., 2010). As such, it would be prudent to test the stability of anhedonia. Therefore, a scale that is able to assess responses “right now” versus “over time” or “in general” would be ideal for measurement in MDD, since they allow for frequent repeat testing. As a result, both the FCPS and SHAPS may be more beneficial in capturing information in the context of a depressive episode. For example, in a sample of inpatients, anhedonia scores based on the FCPS were stable over 7 months despite recovery in two thirds of patients (Clark et al., 1984); however, as inpatients this group may reflect a more chronically ill group. In another naturalistic study in chronic MDD patients, anhedonia (based on the FCPS) did not change over a 1 year follow-up despite reductions in depressive symptoms (Schrader, 1997). Second, the CPAS and CSAS measure pleasure in addition to personality traits, whereas both the FCPS and SHAPS focus exclusively on consummatory pleasure. All four scales incorporate questions relating to both primary and secondary reward; for example, food and sex represent “primary” rewards (inherent rewards), whereas photography or money are secondary rewards (no inherent reward in itself and for which reward value must be learned). Third, both the SHAPS and FCPS exhibit good convergent and discriminant validity: they moderately correlate with depression severity as would be expected, but do not correlate with measures of anxiety (Leventhal et al., 2006; Nakonezny et al., 2010). In contrast, the CPAS has a weak correlation with depression severity, and both the CPAS and CSAS have strong associations with non-affective aspects of personality and psychotic disorders (Leventhal et al., 2006). Fourth, the CPAS, CSAS as well as the FCPS exhibit a high degree of cultural bias, as demonstrated in questions such as “I always find organ music dull and unexciting” (CPAS) or “you are skiing down a mountain very fast while still in good control of yourself” (FCPS). The SHAPS was constructed to avoid cultural bias and so contains items with

wider applicability (“I would be able to enjoy my favourite meal”). As a result, however, one could argue that it does not capture the events or activities that are likely to elicit strong hedonic responses (Lang et al., 2012). The ability to accurately tap into the subjective nature of what individuals find pleasurable or interesting is the greatest challenge in measuring anhedonia.

The development of the Dimensional Anhedonia Rating Scale (DARS) was primarily driven by a need to ensure the assessment of anhedonia with an instrument tapping interest, motivation, effort and pleasure. All four dimensions may be necessary to elicit a complete hedonic response, and a deficiency in any of these dimensions could consequently reduce the experience of pleasure. Neuroscientific evidence suggests that multiple, distinct aspects of reward processing can give rise to loss of pleasure. For example, depressed patients reported similar levels of consummatory pleasure but lower levels of reward anticipation compared to healthy controls (Sherdell et al., 2012). Furthermore, levels of reward anticipation were associated with motivation for effort expenditure only in the participants with MDD (Sherdell et al., 2012). There is evidence that MDD patients have reduced ability to detect reward compared to healthy controls and exhibit reduced brain activity when a reward has occurred, which suggests individuals with MDD do not experience the same level of feedback from rewards as healthy controls (Gradin et al., 2011; Kumar et al., 2008). Taken together, it appears that reward-related deficits in MDD impact much more than consummatory pleasure, and available tasks and tools need to fully reflect all components of reward processing in order to yield an adequate assessment of this cardinal symptom of depression. Inclusion of these elements (desire, motivation, effort and consummatory pleasure) in the DARS is expected to capture a comprehensive representation of the individual anhedonic experience consistent with current neurobiological models.

The development and validation of the DARS was performed in three studies: (1) item selection and preliminary internal reliability testing within a sample of community participants (Study 1), (2) cross-validation of reliability estimates from Study 1 and convergent and discriminant validity assessment in an online study using community participants (Study 2), and (3) reliability and validity testing of the DARS using unipolar or bipolar depressed patients and healthy controls (Study 3).

2. Study 1: Item selection

2.1. Methods

2.1.1. Item generation

The DARS is a self-report scale that measures anhedonia across the following 4 domains: hobbies/past-times, food/drinks, social activities, and sensory experiences. Within each domain, participants are required to provide at least two of their own examples of what they find rewarding (e.g. “gardening, playing the guitar” under “hobbies/past-times”). Illustrative examples are provided within each domain (see Appendix). Subsequently, participants answer a set of standardized questions about desire, motivation, effort and consummatory pleasure for the examples provided.

An item pool comprised of 34 items was created across the domains. The domains were based on results from the factor analysis of focus group data obtained during the SHAPS development (Snaith et al., 1991). Novel items were generated (by SR) within each domain to ensure the presence of at least one item assessing desire, motivation, effort and pleasure. The DARS was designed to be useful in clinical and research settings (i.e. easy to use, minimal participant burden). Consequently, we developed it as a self-report questionnaire with the aim of reducing the item

number to approximately 20 items (~5 min completion time).

The timeframe for the standardized questions was based on their feeling “right now”. This was done in order to be able to gauge state versus trait effects over time. Responses were based on a 5-point Likert scale (Not at all=0; Slightly=1; Moderately=2; Mostly=3; Very Much=4). Scoring was based on summation of all items. Eight items were reverse keyed. A high score reflects increased motivation, effort, and pleasure (i.e., less anhedonia).

2.1.2. Content validity

The 34 item pool was reviewed by five independent, licensed clinical psychologists, who provided qualitative feedback regarding item content (i.e. free-form responses, and a quantitative questionnaire was not utilized). All five reviewers endorsed the face validity of items, made comments to clarify the concepts addressed, and wording of questions were suggested. The scale was revised according to comments.

2.1.3. Participants

Community participants were recruited from advertisements in local media. Participants were required to meet the following inclusion criteria: age between 18 and 65 years, at least 8 years of education, and capacity to provide informed consent.

2.1.4. Procedures

Participants attended two research appointments at the Centre for Addiction and Mental Health (CAMH) as part of a large community-based study that included other measures than the DARS. Following written informed consent, the Structured Clinical Interview for Axis I DSM-IV Disorders (SCID-I) (First et al., 2002) and the Dimensional Interview for Personality Disorders (DIPD), were used to evaluate the presence of psychiatric disorders or personality disorders, respectively. Demographic information was obtained and the participants subsequently completed the DARS. Participants received monetary compensation in return for participation.

2.1.5. Statistical analysis

The structure of the DARS was evaluated using principal components analysis (PCA) with a promax rotation, using parallel analysis to determine the number of components to extract (Hayton et al., 2004). Items with communalities less than 0.5 were removed (Field, 2005). Loadings with absolute values ≥ 0.5 were considered to contribute sufficiently to the overall variability accounted for by the component (Stevens, 1992; Costello and Osborne, 2005); items with loadings > 0.3 on more than one component were removed to improve consistency.

Internal consistency reliability for the items in the full scale and any subscales retained in the DARS was conducted using Cronbach's α . An α above 0.7 was considered acceptable (Devellis, 2012). Other reliability tests including the average inter-item correlation (AIC) were also performed. Group differences based on gender, ethnicity, and depression treatment were evaluated with student *t*-tests or Mann–Whitney U tests depending on the normality of the data. Where there was more than one test of the same type performed for a given hypothesis, Bonferroni correction for multiple comparisons was utilized.

2.2. Results

2.2.1. Subject characteristics

A total of 229 participants recruited from the community were enrolled in this study. The demographic data represent a middle-aged sample, most of whom completed at least high school (Table 1). Overall, a third of the sample had a current psychiatric diagnosis of which 21% met criteria for MDD in the past month.

Table 1
Demographic information of Study 1 and 2 Participants.

	Study 1 Item selection	Study 2 Cross-validation
Enrolled	229	150
% Female	45.4	70.6
Age (SD)	41.7 (12.7)	27.5 (11.9)
Ethnorace(%)		
Caucasian	57.6	74.8
African	11.8	6.8
Hispanic	3.9	6.1
Native	1.3	4.1
Asian	6.1	2.7
East Indian	6.1	
Pacific Islander	2.6	
Bi-racial	6.6	1.4
Other	3.9	4.1
Education	74.6% > High school	75.3% > High school
% Born in Canada	62.9	–
% Employed	58.1	–
Family history–Mental illness (%)	32.3	–
Current axis I diagnosis (%)	33.2	31.0*
Major depression	21.1	–
Bipolar disorder	1.3	–
Schizophrenia	7.9	–
OCD	6.5	–
Panic disorder	5.3	–
PTSD	6.5	–
GAD	21.1	–
Alcohol/Substance use disorder	19.7	–
Eating disorder	2.6	–

* Based on self-reported lifetime treatment for Major Depressive Disorder.

2.2.2. Scale structure and item reduction

The parallel analysis of the 34-item DARS yielded a 5-component solution ($\lambda_1=12.04$, $\lambda_2=3.46$, $\lambda_3=2.84$, $\lambda_4=1.77$, $\lambda_5=1.54$). Specifying 5 components in the PCA yielded communalities ranging from 0.249 to 0.821. The rotated loadings revealed that items were grouped according to social activities (35.4% variance explained), food/drinks (10.2% variance explained), hobbies/pastimes (8.4% variance explained), sensory experience (5.2% variance explained), and one component that was comprised of only the negatively keyed items (4.5% variance explained). The intercorrelation among components ranged from 0.45 to 0.63.

Based on the criteria set out for item reduction, 7 items with communalities below 0.5 were removed (Field, 2005). Since the reverse key items across domains strongly loaded onto only one component with little or no correlation within or across components, the 3 remaining of these questions were removed. The items left all had high loadings (≥ 0.5) on their respective component. The next step in item reduction was to ensure that one to two items assessing desire, motivation, effort and pleasure facets were included within each domain. The decision on which items within a facet to retain was primarily based on component loadings. Of note, the motivation item within the hobbies domain was reverse-keyed item and removed earlier; therefore, this domain did not include an item assessing motivation. This process yielded a 17-item scale comprised of four components mapping onto the different domains of anhedonia: hobbies, foods/drinks, social activities and sensory experience (Appendix 1). The highest score attainable is a 68, where a high score is reflective of less anhedonia.

2.2.3. Themes of self-report examples

Self-report examples were reflective of their respective domains, indicating an understanding of the categories. Examples were reviewed to elicit themes. Overall themes for hobbies and

social activities were: arts and crafts, lifestyle/culture, leisure, fitness/wellness/sports, multimedia/technology, education/training, food/cuisine, home and garden. The most frequent hobbies reported were reading and a form of exercise, while for social activities meeting friends/family for dinner or watching movies/TV with others were the most common examples. There was an overlap in examples between hobby and social domains in concept, however, within the social domain it was specified the activity was with other people and not solitary (e.g. watching hockey in hobbies vs. playing hockey with friends in social activities).

2.2.4. Reliability analysis

The Cronbach's α was 0.92 for the DARS total score. The Cronbach's α for the domains were also high: pastimes/hobbies (0.91), foods/drinks (0.86), social activities (0.83), and sensory experiences (0.89). The AIC was 0.41 for the whole scale and ranged between 0.59 and 0.72 for the subscales (hobbies: 0.72; foods/drinks: 0.63; social activities: 0.59; sensory experiences: 0.63).

2.2.5. Association of other variables with DARS scores

The mean DARS total score for the whole sample was 55.8 ± 10.6 with the subscale means as follows: hobbies (13.9 ± 3.0), food/drink (12.8 ± 3.4), social (12.3 ± 3.3), and sensory (16.8 ± 4.0). These values did not vary by ethnicity or gender. However, those with current MDD had lower scores than participants with no psychiatric history (51.2 ± 10.1 vs. 57.5 ± 9.0 , $p < 0.012$). Among the domains, this group difference was observed within the social subscale with a trend toward significance in the sensory subscale (social: 10.0 ± 3.1 vs. 12.9 ± 2.7 , $p = 0.001$; sensory: 15.3 ± 4.0 vs. 17.4 ± 3.5 , $p = 0.016$, corrected for multiple comparisons).

2.3. Discussion

The DARS demonstrated excellent internal consistency reliability, as evidenced by the high alphas overall and in the subscales. The majority of examples provided accurately reflected the category definitions. Based on the initial item selection study, less than 1% of total responses ($N = 1,832$ across categories) across categories were not reflective of the domain.

The initial item pool exhibited a five-component structure, reflecting the domains of hobbies, foods/drinks, social activities, sensory experience, as well as a component that included all of the reverse-keyed items (negatively valenced wording). Scale development techniques for item development recommend the inclusion of items that are reversed in order to avoid “agreement bias” – the tendency to agree with items irrespective of their content (Devellis, 2012). While this method may reduce acquiescent response styles, the disadvantages of including items worded in the opposite direction may outweigh these potential benefits: commonly, reverse-keyed items perform poorly psychometrically (Currey et al., 2002). The separation of reverse keyed items in a component rather than a substantive feature of anhedonia is consistent with this observation. The fifth component accounted for less than 5% of the variance attributed to the full structure. Additionally, the items within the component did not strongly correlate to the respective component. Five of eight items in this component assessed “motivation”, suggesting that this component may be reflective of this reward facet rather than reverse polarity. In either case, the poor item performance necessitated their exclusion.

The substantive components reflected the domains of anhedonia as opposed to the components (interest, motivation, effort, consummatory pleasure). This suggests that reward type is a significant factor when measuring anhedonia. The distinction of

reward-type in the DARS is also consistent with the notion of primary versus secondary reward. Evidence supports both overlapping and distinct neurobiological response to primary and secondary reward (Sescousse et al., 2010, 2013), and incentive-type specific activation within reward processing. Consequently, the DARS is well-positioned to test the neurobiological correlates of subjective hedonic response across different reward types.

The anhedonia facets (interest, motivation, effort and pleasure) may not have been distinct enough from each other in order to be distinguished as separate components. In other words, the nuance among these facets may be too subjectively subtle. A larger item set size may also be necessary to capture components of anhedonia as opposed to type. For example the Temporal Experience of Pleasure Scale is an 18 item questionnaire to evaluate anticipatory and consummatory pleasure in the context of physical pleasure (Gard et al., 2006). Indeed the factor analysis did reveal a two-factor structure consistent with this distinction. In contrast, the recently published 17-item Anticipatory and Consummatory Interpersonal Pleasure Scale (ACIPS) was also designed to distinguish between anticipation and consummatory pleasure within a social domain, but the factor solution mapped onto type of social experiences (Gooding and Pflum, 2014). This may be due to inherent scale characteristics, or that social pleasure encompasses both anticipation and consummatory pleasure concurrently, so the distinction may be difficult to ascertain on a self-report measure.

Finally, convergent and divergent validity estimates were not assessed in this study. Deriving these data as well as cross-validating the internal reliability estimates from Study 1 in a community sample were the objectives of Study 2.

3. Study 2: Cross-validation

3.1. Methods

3.1.1. Subject sample

Recruitment for this study was through the “online laboratory” of the Hanover College Psychology Department website (<http://psych.hanover.edu/research/exponnet>) that lists active online studies in the area of Psychology. Evidence indicates that the psychometric properties of scales performed in a web-based vs. laboratory-based environment may be comparable (Risko et al., 2006). Therefore, web-based scale administration provides an opportunity to assess the psychometric characteristics of a scale in a cost-effective way.

3.1.2. Procedures

Participation was anonymous. Community participants between the ages of 18–60 were enrolled. No other inclusion criteria were added in order to obtain a broad sample of subjects with a range of anhedonia for the purpose of confirming item selection in Study 1. Upon clicking the website link for the DARS, participants were directed to a secure external website affiliated with the Department of Psychiatry, University Health Network (UHN). Interested participants provided consent and completed an online package of questionnaires.

3.1.3. Measures

The questionnaire package included the following measures to evaluate convergent and divergent validity. Convergent validity was assessed with the current gold standard anhedonia scale, SHAPS. The SHAPS is a 14-item self-assessment scale estimating the degree to which a person is able to experience pleasure or to anticipate a pleasurable event (i.e. hedonic tone) (Snaith et al., 1995). A score of 2 or more “disagree/definitely disagree” is considered to be indicative of an anhedonic state (i.e. higher score

reflects greater anhedonia). The SHAPS was developed specifically for use in MDD.

Convergent validity was also assessed using the Behavioral Inhibition System and the Behavioral Activation System Scales (BIS/BAS), a 20-item self-report measure of punishment sensitivity (BIS subscale) as well as drive, motivation and reward responsivity (BAS subscales of drive, reward response and fun seeking) (Carver and White, 1994). Psychometric properties of the BIS/BAS Scales reflect convergent validity with depression and anxiety measures (Campbell-Sills et al., 2004). The subscales of the BAS were used to establish convergent validity.

Divergent validity was assessed using measures of depression severity, physical activity, and behavioral inhibition. The Centre for Epidemiological Studies in Depression (CESD) is a 20-item self-report scale of depressive symptomatology, and has been used as a basic screening tool for depression (Radloff, 1977). A score greater than 16 is indicative of potential clinical depression. Physical activity was assessed using the NASA Activity Scale (NASA), a 1 item physical activity scale to assess the degree of physical activity over the preceding month (Wier et al., 2001). The BIS subscale from the BIS/BAS Scales was also used to evaluate divergent validity as it measures punishment and harm avoidance, and inhibition of reward seeking.

3.1.4. Statistical analysis

Internal consistency for the items in the full scale and subscales was conducted using Cronbach's α and the AIC. Convergent and divergent validity were established using correlations between total and subscale scores of the DARS and the SHAPS, BIS/BAS, CESD or NASA physical activity scale. Either Pearson's correlation coefficients or Spearman's Rank (r_s) correlations were conducted based on whether the variables followed a normal distribution. Group differences based on gender, ethnicity, and depression treatment were evaluated with student t -tests or Mann–Whitney U tests depending on the normality of the data. Where there was more than one test of the same type performed for a given hypothesis, Bonferroni correction for multiple comparisons was utilized.

3.2. Results

3.2.1. Subject characteristics

A total of 150 participants completed the online study. Demographic characteristics are presented in Table 1. Those who reported treatment for MDD had higher CESD scores than participants with no psychiatric history (25.2 vs. 18.4, $p=0.001$). Self-report themes were similar to Study 1.

3.2.2. Reliability analysis

The Cronbach's α was 0.92 for the DARS total score. The Cronbach's α for the domains were high, similar to Study 1 findings: pastimes/hobbies (0.90), foods/drinks (0.75), social activities (0.85), and sensory experiences (0.89). The AIC for the total scale was 0.41, while the AICs for the subscales ranged from 0.46 to 0.69

Table 2

Reliability assessment of the DARS across Study 1 and 2.

	Cronbach's alpha (α)		Inter-item correlation	
	Study 1	Study 2	Study 1	Study 2
Total scale	0.92	0.92	0.41	0.41
Pastimes/Hobbies	0.91	0.90	0.72	0.69
Foods/Drinks	0.86	0.75	0.63	0.46
Social activities	0.83	0.85	0.59	0.59
Sensory experiences	0.89	0.89	0.63	0.63

Table 3

Validity of the DARS total score and subscales in Study 1-online sample.

DARS	SHAPS	BAS-reward	BAS-fun	BAS-drive	CESD
Total scale	–0.58**	0.53**	0.18*	0.35*	–0.47**
Pastimes/Hobbies	–0.47**	0.44**	0.12	0.26*	–0.47**
Foods/Drinks	–0.36**	0.31**	0.05	0.17*	–0.29**
Social Activities	–0.57**	0.51**	0.35**	0.45*	–0.47**
Sensory Experiences	–0.45**	0.46**	0.13	0.32*	–0.31**

Boldface type represents moderate and strong significant correlations.

BAS: Behavioral Activation System; CESD: Centre for Epidemiological Studies in Depression; DARS: Dimensional Anhedonia Rating Scale; SHAPS: Snaith–Hamilton Pleasure Scale

* $p < 0.05$.

** $p < 0.001$.

(Table 2).

3.2.3. Validity analysis

The DARS total score demonstrated strong convergent validity with the SHAPS ($r_s = -0.58$, $p < 0.001$), as well as a strong correlation with BAS-reward ($r_s = 0.53$), moderate correlation with BAS-drive ($r_s = 0.35$, $p < 0.001$) and a small correlation with BAS-fun ($r_s = 0.18$, $p = 0.034$). The DARS subscales all had moderate correlations with the SHAPS and BAS-reward, and small to moderate correlations with BAS-drive (Table 3). The DARS-social subscale was the only domain to have a correlation with BAS-fun ($r_s = 0.35$, $p < 0.001$).

Divergent validity was demonstrated by the lack of correlation between the DARS total score and subscales with physical activity and the BIS. In addition there was a medium correlation with the CESD total score ($r_s = -0.47$, $p < 0.001$). Similar correlations between the CESD and hobbies ($r_s = 0.47$, $p < 0.001$), social subscale (both $r_s = -0.47$, $p < 0.001$) were observed and small correlations were demonstrated for the foods/drinks ($r_s = -0.29$, $p < 0.001$) and moderate with the sensory experience subscale ($r_s = -0.31$, $p < 0.001$).

3.2.4. Association of other variables with DARS scores

The mean DARS score was 50.0 ± 12.6 , with the subscale means as follows: hobbies (12.7 ± 3.6), food/drink (10.9 ± 3.4), social (10.8 ± 4.0), and sensory (15.3 ± 4.6). In addition, the total score did not vary by ethnicity or gender. Those who reported treatment for MDD had lower DARS total scores (46.1 vs. 51.2, $p < 0.01$). Among the subscales, a group difference was only observed within the DARS social domain, where the MDD group had lower scores (9.3 vs. 11.6, $p = 0.005$).

3.3. Discussion

Similar to Study 1, the DARS demonstrated high internal consistency. Convergent validity was demonstrated through the strong correlation with an existing measure of anhedonia. Yet, this strong correlation is far from unity, which is anticipated given that the content domain of the DARS extends well beyond consummatory pleasure. Convergent validity was also supported by moderate correlations between the DARS total score and depression severity based on the CESD ($r_s = -0.37$ – 0.47), which is consistent with the diagnostic criteria for MDD as well as previous reports describing a similar correlation between depression and the SHAPS (Snaith et al., 1995; Franken et al., 2007; Leventhal et al., 2006; Nokonezny et al. 2011). The lack of correlation with the BIS subscale is in line with expectations, as aversive motivation is operationalized as orthogonal to appetitive motivation in this measure, with distinct neurobiological substrates (Carver and White, 1994). Finally, divergent validity of the DARS with physical

activity was demonstrated by the lack of correlation with the NASA.

The DARS did not differ by age, or gender, which provides support for its use in various populations. In addition, the evidence suggests a lack of difference by ethnicity, but this needs to be more rigorously tested in diverse samples in order to confirm this initial finding. The SHAPS was designed, in part, to avoid the generalizability issues in other anhedonia scales (Nakonezny et al., 2011; Franken et al., 2007; Nagayama et al. 2012; Liu et al., 2012; Chapman, 1979; Fawcett et al., 1983). The issue that arises with increased generalizability is the lack of specificity in items that elicit a strong hedonic response. To this effect, the DARS has demonstrated generalizability while retaining item specificity.

Importantly, while the evidence based on the scale score averages and reliability estimates suggest similarity across online and paper DARS, whether both assessment-types yield truly comparable results would require testing both formats in a single study.

Considering the DARS was developed for use in depression, the objective for Study 3, then, was to determine internal reliability as well as convergent and divergent validity in MDD and bipolar patients in a depressive episode.

4. Study 3: Validation in MDD

4.1. Methods

4.1.1. Subject sample

Recruitment for this study was through referral from the Department of Psychiatry at UHN, participant flyers at UHN and CAMH, as well as the research participant database at CAMH. All participants were required to be between the ages of 18–65 with no significant/unstable medical illness. Further inclusion criteria for healthy controls were: no psychiatric history, including alcohol or substance abuse/dependence, no lifetime use of psychotropic medication. Additional inclusion criteria for the MDD group were: presence of an Axis I diagnosis of MDD or Bipolar Disorder, meeting DSM-IV criteria for a current Major Depressive Episode (MDE) as defined by the Mini International Neuropsychiatric Interview (MINI) (Sheehan et al., 1998), no other primary Axis I or II disorders, and no substance abuse or dependence in the last 6 months.

4.1.2. Measures

The same measures as in Study 2 were used to assess the validity of the DARS: SHAPS, BIS/BAS, CESD and NASA.

4.1.3. Procedures

One research visit to UHN was required. Following written informed consent, demographic information was collected and all patients and healthy controls underwent the MINI (Sheehan et al., 1998) to determine the presence of Axis I psychiatric disorders according to DSM-IV criteria. Subsequently, the DARS and additional questionnaires were administered, as specified above. In addition to the scale battery, a feedback questionnaire was administered to all participants with 4 quantitative questions answered on a 5-point Likert scale (“Were the instructions at the beginning on how to complete the scale clear?: not at all (1) – very clear (5); “Were the questions understandable and clearly worded”: not at all (1) – very clear (5); “Overall, how easy was the scale to complete?”: Difficult (1) – easy (5); “What do you think about how long it takes to complete the scale?”: very long (1) – very short (5)) and a qualitative question (“Was there anything else about completing the scale that you did not understand?”).

4.1.4. Statistical analysis

Internal consistency reliability for the items in the full scale and any subscales retained in the DARS was conducted using Cronbach's α and AIC. Convergent and divergent validity were established using correlations between total scores of the DARS and the SHAPS, BIS/BAS Scales, CESD or NASA physical activity scale. Either Pearson's correlation coefficients or Spearman's Rank (r_s) correlations were conducted, based on whether the variables followed a normal distribution. Group differences based on gender ethnicity, and depression treatment were evaluated with student t-tests or Mann–Whitney U tests depending on the normality of the data. Bonferroni correction for multiple comparisons was utilized.

To ascertain the incremental value of the DARS over the SHAPS, a hierarchical regression analysis was conducted with the SHAPS entered as the first independent variable, followed by the DARS in the second step. Four linear regressions were run where in each the dependent variable was, respectively, depression severity (CESD), and behavioral activation (BAS subscales drive, reward, and fun). Two logistic regressions were also performed, where the dependent variables were, respectively, depression status (MDD vs. HC) as well as treatment resistant depression (TRD) status within the depressed group (TRD vs. non-TRD).

4.2. Results

4.2.1. Subject characteristics

A total of 102 participants, including adults experiencing an MDE ($n=52$) in the context of MDD ($n=44$) or bipolar disorder ($n=8$), as well as healthy controls ($n=50$) were enrolled in the study. Among the 44 MDD patients, 15 were considered treatment resistant, defined as a failure of at least 4 antidepressants for the current episode. Age and gender were not different between groups, although the depressed group was less ethnically diverse (Table 4). While both groups reported high years of education, reflecting the completion of post-secondary education, the healthy controls had significantly more years (18.1 vs. 16.1, $p=0.004$). As expected based on previous literature, Axis III comorbidity was higher in the MDE group compared to the control group.

Table 4
Demographic information in depressed patients and healthy controls.

Variable	Depressed ($n=52$)	Healthy controls ($n=50$)	p
Age (years)	43.4 (14.3)	37.8 (13.1)	0.065
Gender (% female)	61.5%	74.0%	0.208
Education (years)	16.1 (2.8)	18.1 (3.1)	0.004
Ethnorace (%)			
Caucasian	84.6	50.0	0.002
African	1.9	2.0	
Hispanic	1.9	0	
East Indian	5.7	16	
Asian	0	26.0	
Pacific Islander	1.9	2.0	
Other	3.8	4.0	
Axis III diagnosis (%)	32.7	16.0	0.05
Hypertension	26	12.5	
Diabetes	21.1	0	
Thyroid dysfunction	21.1	25	
High cholesterol	15.8	0	
Arthritis	10.5	0	
Chronic pain	15.8	0	
Osteopenia	0	12.5	
CESD total score	39.5 (9.9)	6.3 (7.1)	< 0.001
Duration of MDE (months)	60.8 (73.6)	n/a	n/a

Bold highlights all of the tests that were statistically significant.

Table 5
Reliability and validity of the DARS total score and subscales in MDD sample.

DARS	Reliability		Convergent and divergent validity ^a				
	Cronbach's alpha (α)	Inter-item Correlation	SHAPS	BAS-reward	BAS-fun	BAS-Drive	CESD
Total scale	0.96	0.60	–0.79	0.48	0.51	0.42	–0.37
Pastimes/Hobbies	0.91	0.73	–0.78	0.41	0.37	0.35	–0.36
Foods/Drinks	0.90	0.70	–0.63	0.52	0.44	0.42	–0.34
Social activities	0.88	0.66	–0.73	0.44	0.49	0.42	–0.30
Sensory experiences	0.94	0.75	–0.67	0.32	0.46	0.32	–0.29

BAS: Behavioral Activation System; CESD: Centre for Epidemiological Studies in Depression; DARS: Dimensional Anhedonia Rating Scale; MDD: Major Depressive Disorder; SHAPS: Snaith–Hamilton Pleasure Scale

^a All convergent and divergent correlations significant at $p < 0.05$.

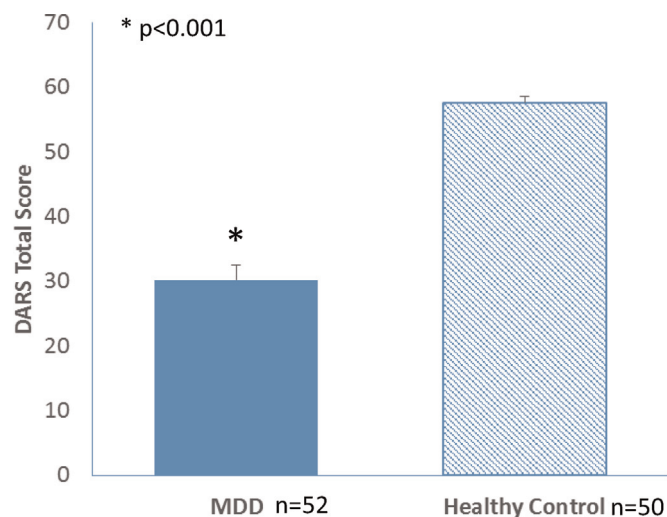


Fig. 1. Difference in DARS-17 score between MDD and Healthy Controls.

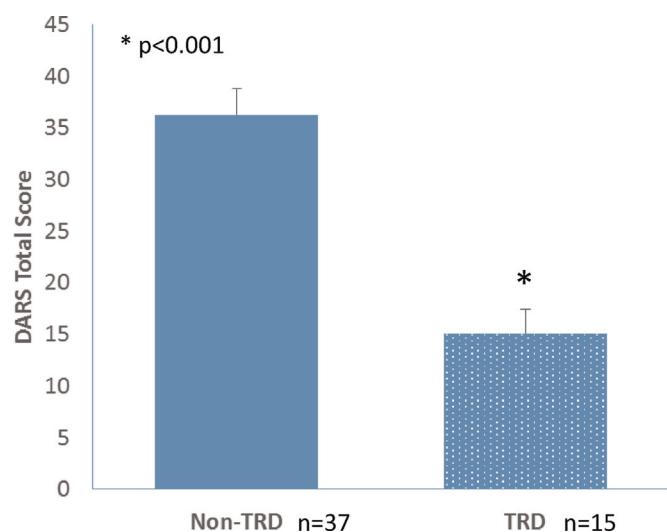


Fig. 2. DARS and SHAPS scores in TRD and non-TRD patients.

4.2.2. Reliability analysis

The Cronbach's α was high at 0.96 for the DARS total score and also high for the subscales: pastimes/hobbies (0.91), foods/drinks (0.90), social activities (0.88), and sensory experiences (0.99). The AIC for total score was 0.60, while the subscales ranged from 0.66 to 0.75 (Table 5).

4.2.3. Validity analysis

The DARS total score demonstrated high convergent validity with the SHAPS ($r_s=0.79$, $p < 0.001$) as well as moderate

Table 6
Differences in reward between TRD and non-TRD patients.

Scale	Non-TRD (n=37)	TRD (n=15)	p-Value
DARS			< 0.001
Total	36.3 (15.5)	15.1 (8.8)	
Hobbies	8.6 (4.4)	3.7 (2.7)	
Food/drink	10.0 (4.2)	4.7 (2.9)	
Social	7.1 (3.9)	2.7 (2.3)	
Sensory	10.7 (5.6)	3.9 (2.4)	
SHAPS	10.2 (2.7)	5.3 (3.6)	< 0.001
BAS-reward	14.8 (2.6)	11.2 (3.8)	0.001
BAS-drive	6.9 (2.6)	9.4 (2.7)	0.003
BAS-fun	10.8 (2.3)	7.1 (2.7)	< 0.001
BIS	25.1 (7.5)	25.1 (3.3)	ns

BAS: Behavioral Activation System; BIS: Behavioral Inhibition System; DARS: Dimensional Anhedonia Rating Scale; SHAPS: Snaith–Hamilton Pleasure Scale; TRD: Treatment Resistant Depression

correlations with the BAS-Drive ($r_s=0.42$, $p=0.002$), BAS-Reward ($r_s=0.48$, $p < 0.001$), and BAS-Fun ($r_s=0.52$, $p < 0.001$). Anhedonia based on the DARS was not related to behavioral inhibition, and demonstrated a moderate correlation with depression severity ($r_s=0.37$, $p=0.036$) and physical activity ($r_s=0.4$, $p=0.004$). All DARS subscales had moderate correlations with the SHAPS, BAS and small to moderate correlations with the CESD (Table 5).

4.2.4. Group differences

No differences based on age, gender, or ethnicity were observed for the DARS total or subscale scores. Years of education did not correlate with DARS scores within the MDE or healthy control group. Within the MDE group, there were also no differences between the bipolar and unipolar depressed patients on the DARS total or subscale scores. However, the MDE group had a lower DARS total score than healthy controls (30.2 vs. 57.6, $p < 0.001$) (Fig. 1). Based on frequency distributions, MDE participants had a score distribution of 1–44, whereas healthy controls had a distribution of 44–68. Similarly, the MDE group had lower scores for all of the subscales: hobbies (7.1 vs. 14.9, $p < 0.001$), foods/drinks (8.5 vs. 13.3, $p < 0.001$), social activities (5.8 vs. 12.8, $p < 0.001$), and sensory experience (8.7 vs. 16.7, $p < 0.001$). The subanalysis of TRD patients demonstrated lower levels of anhedonia based on the DARS (Fig. 2) and SHAPS as well as behavioral approach than non-resistant MDE patients, while behavioral inhibition was not significantly different (Table 6).

4.2.5. Hierarchical regression analysis

The hierarchical analysis was done using scale variables (e.g. CESD and BIS/BAS) as well as categorical variables (MDE vs. healthy controls; TRD vs. healthy controls). Multicollinearity indices of tolerance and variance inflation factor were above 0.1 and less than 10, respectively. Depression severity was predicted by the SHAPS ($\beta=2.23$, $p < 0.001$), while the overall model fit was

Table A1

The Dimensional Anhedonia Rating Scale.

	Not at all	Slightly	Moderately	Mostly	Very much
A. Please list at least 2 of your favorite pastimes/hobbies that are NOT primarily social:					
1. I would enjoy these activities					
2. I would spend time doing these activities					
3. I want to do these activities					
4. These activities would interest me					
B. Please list at least 2 of your favorite foods or drinks:					
5. I would make an effort to get/make these foods/drinks					
6. I would enjoy these foods/drinks					
7. I want to have these foods/drinks					
8. I would eat as much of these foods as I could					
C. Please list at least 2 of your favorite social activities:					
9. Spending time doing these things would make me happy					
10. I would be interested in doing things that involve other people					
11. I would be the one to plan these activities					
12. I would actively participate in these social activities					
D. Please list at least 2 of your favorite sensory experiences:					
13. I would actively seek out these experiences					
14. I get excited thinking about these experiences					
15. If I were to have these experiences I would savor every moment					
16. I want to have these experiences					
17. I would make an effort to spend time having these experiences					

$R^2=0.62$. Only an additional 2.6% of the variance explained when the DARS was entered into the model ($R^2=0.64$, $p=0.01$; $\beta=-0.32$, $p=0.011$). For all of the BAS subscales (drive, reward, fun), the SHAPS was significant as a single predictor ($\beta=-0.34$, $p<0.001$ and $R^2=0.23$, $p<0.001$; $\beta=-0.53$, $p<0.001$ and $R^2=0.45$, $p<0.001$; and $\beta=-0.38$, $p<0.001$ and $R^2=0.31$, $p<0.001$). For all the BAS subscales, the additional variance explained by the DARS was 14.1% ($R^2=0.36$, $p<0.001$; $\beta=0.12$, $p<0.001$), 5.7% ($R^2=.50$, $p=0.001$; $\beta=-0.09$, $p=0.001$), and 7.1% ($R^2=0.38$, $p=0.001$; $\beta=0.08$, $p=0.001$).

Using logistic regression to predict group status (MDE vs. healthy control), both the SHAPS and DARS equally performed, with the SHAPS explaining 51.9% of the variance ($\chi^2=73.11$, $p<0.001$). The model improved to 56.2% when the DARS was included ($\chi^2=9.37$, $p<0.001$). In predicting TRD status, the SHAPS on its own accounted for 31.1% of the variance ($\chi^2=19.01$, $p<0.001$). When both the SHAPS and DARS were entered, the model explained 38.8% of the variance ($\chi^2=6.01$, $p<0.001$), however the SHAPS was no longer a significant predictor when the DARS was entered into the model.

4.2.6. Participant feedback

Over 90% of participants rated the DARS feedback questions with a 4 or higher, indicating excellent clarity of the DARS instructions, questions, and ease of use. Over 70% of participants felt the time to complete the scale was short. MDE patients were statistically more likely to rate the clarity of instructions lower, however the means were almost the same (4.6 vs. 5.0, $p=0.003$). There were no other group differences with respect to feedback. Qualitative comments offered by participants fell into four themes related to: understanding of the domains, difficulty producing self-report examples, question content, and Likert scale content. The most prominent comment was related to the difficulty in coming up with examples.

4.3. Discussion

Across the studies, the DARS consistently demonstrated excellent internal consistency reliability, as evidenced by the high alphas overall (0.91–0.96) and in the subscales (0.75–0.92).

The DARS demonstrated a strong correlation with the SHAPS in

the MDD sample (Study 3) and in the online sample (Study 2). A high correlation between the DARS and SHAPS in the MDD sample could reflect a general dampening of reward responsivity across components of anhedonia, resulting in similar responding across interest, motivation, effort and pleasure. Support for this hypothesis comes from the DARS correlations with the BAS, which measures trait reward responsiveness (Carver and White, 1994). The subscales of fun (sensation seeking), drive (goal-directed reward behavior) and reward (responsivity to reward) are also indicators of reward motivation and responsiveness (Quilty et al., 2014). The correlations of the DARS across the subscales were moderate but much stronger in the MDD trial than in the online trial, particularly for the fun and drive subscales. For example, in the online sample, the correlation of the DARS with BAS-fun was small but significant ($r_s=0.18$, $p=0.03$), whereas in the MDD sample this correlation was moderate and significant ($r_s=0.51$, $p=0.001$). The correlations with the BAS subscales also provide support for the multifaceted nature of the items in the DARS.

Across both Studies 2 and 3, there were significant moderate correlations between the DARS total score and depression severity based on the CESD ($r_s=-0.37$ – 0.47) consistent with previous reports of the SHAPS (Snaith et al., 1995; Franken et al., 2007; Leventhal et al., 2006; Nokonezny et al., 2011). Interestingly, in the MDD sample, the CESD correlations with the DARS total and subscales were lower than in the online sample, which also lends support to the idea that anhedonia represents an overlapping but distinct construct with depression. The lack of correlation between CESD and DARS scores in the TRD only group also suggests that anhedonia is a distinct construct. However, a lack of correlation based on greater restriction in range of scores cannot be ruled out. Similar to the online sample in Study 2, there was an expected lack of correlation with the BIS. In addition, there was a moderate correlation between the DARS and physical activity in the MDD sample ($r_s=0.4$, $p=0.004$), which was not observed in the online study. In one vein, this is not surprising given lack of energy is a diagnostic feature of an MDE (DSM-5, 2013) and associations between anhedonia and low energy have been reported (Brown et al., 1995). Finally, the DARS did not differ by age, gender or ethnicity across all studies, which again, supports its use across groups. However, it should be noted that there were differences between the MDD and control group in ethnicity representation,

Table A2
Examples of Activities/Experiences for DARS categories.

Hobbies									
Arts and crafts			Lifestyle/ Culture	Leisure/Games	Fitness/Wellness/ Sports	Multimedia/ Technology	Education/Training	Food/cuisine	Home/garden
BeadingActingSketchingSoap makingKnittingPlay guitarWriting stories			● Concerts ● Museums ● Theater ● Dance shows ● Comedy clubs ● Listen to music	● Reading ● Boating ● Shopping ● Travel ● Play with pet ● Nature walks ● Puzzles	● Gym workout ● Spinning ● Biking ● Martial arts ● Running ● Swimming ● Yoga ● Meditation ● Skiing ● Skating	● Watch movies/tv ● Videogames ● Internet surfing ● Download music ● Programming	● Dog training ● Equine therapy ● Science research	● Cooking ● Baking	● Gardening ● Yard work ● House repair ● Cleaning
Social activities									
Arts and crafts	Lifestyle/ Culture	Leisure/Games	Fitness/Wellness/ Sports	Multimedia/ Technology	Education/ Training	Food/cuisine			
● Play in band ● Act in play	● Going dancing ● Going to concerts ● House party ● Going to plays ● Dance classes ● Art exhibits ● Meet new people	● Bowling ● Cards ● Chess ● Shopping ● Playing pool ● Cottage trips ● Camping ● Karaoke ● Play with grandkids	● Biking ● Going to gym ● Going to sporting event ● Hockey	● Watch movies/tv ● Emailing ● Chatroom ● Facebook	● Teaching genealogy ● Volunteering	● Cooking with partner ● Dinner with friends/family ● Coffee with friends/family ● Cooking classes			
Foods and drink									
Foods		Drinks							
● Bagels ● Brownies ● Butter chicken ● Dumplings ● Chicken noodle soup ● Lasagna ● NY strip steak ● Perogies ● Sushi Sensory experience		● Bubble tea ● Coffee ● Gatorade ● Fruit smoothies ● Margaritas ● Red wine ● Coca-cola ● Chocolate milk ● Orange juice							
Smell		Touch	Hearing		Sight		Other		
● Bread ● Nature ● Home cooked food ● Partner's fragrance ● Candles ● Sea ● Coffee ● Flowers ● Hawaiian pizza		● Massage ● Sex ● Feel sun on skin ● Cuddling ● Back scratch ● Petting dog ● Hot bath ● Hugs	● Listening to music ● Comedy podcasts ● Waterfalls ● Sports on radio ● Grandson ● Cello music in garden		● Sunrise ● Sunset ● Viewing art ● People watching ● Star gazing ● Daughter sleeping ● Fireworks ● Waves		● Being by water ● Laying on beach ● Waves crashing on rocks ● Drive up north ● Hot beverage ● Running in the sun ● Swimming in sea ● Mindfulness meditation		

particularly within the Asian group, which constituted 26% of the control group and 0% of the MDD group. As such, whether the DARS truly differs by ethnicity would need to be tested in a study designed to address this question to be confirmed.

The MDD sample demonstrated a 27-point difference from the healthy controls, and likewise, the TRD group demonstrated a 17-point difference from the non-TRD group. These findings suggest the potential sensitivity of the DARS in distinguishing groups, including depressive subtypes. It also demonstrates a significant association of anhedonia with TRD. TRD is characterized by high levels of anhedonia, as well as psychomotor retardation (Malhi and Berk, 2007). Both symptoms are strongly affected by dopaminergic fluctuations (Dunlop and Nemeroff, 2007), which supports the role of a dopaminergic deficit in TRD.

In Study 3, both the SHAPS and the DARS were similarly able to distinguish between healthy controls and a depressed group, with the DARS only contributing an additional 4% improvement in the model. A similar finding was observed for prediction of CESD total scores. Where the two scales differed was in the ability to predict BAS subscales and MDD subtypes. While the SHAPS did have predictive value for all of the BAS subscales, the introduction of the DARS into the model, rendered the SHAPS an insignificant unique contributor. This was particularly evident for the drive subscale where the DARS contributed to an additional 14% to the model, which demonstrates the DARS is superior at estimating levels of motivation than the SHAPS. A similar occurrence was observed when predicting TRD status, where the SHAPS was no longer significant when the DARS was entered into the model. The ability to define MDD subgroups clinically and linking such a clinical discriminator to neurobiology is particularly relevant in order to improve strategies for treatment selection (Kennedy et al., 2012).

5. General discussion

The DARS reflects a refined measurement of anhedonia by including aspects of interest, motivation, effort and consummatory pleasure across reward domains, with additional predictive utility over the SHAPS, and may be able to identify MDD subtypes (i.e. TRD). The DARS demonstrated good internal consistency reliability as well as convergent and divergent validity.

The subscale differences across studies in the “depressed” groups are also revealing. Social anhedonia was the most prominent dysfunction in Studies 1 and 2, whereas all domains were affected in the participants with MDD in Study 3. However, research suggests that social anhedonia declines with treatment in depression (Blanchard et al. 2001). Nevertheless, considering that the MDD sample in Study 3 was more severely depressed than in the online study (mean CESD scores of 18 vs. 40, where above 16 is the cutoff for clinical depression), this finding suggests that at lower severities, social anhedonia is still present.

Several limitations across the DARS studies should be noted. First, item selection was based entirely upon internal consistency and structural analyses with no validity assessments. Second, the cross-validation study was conducted online, precluding a fulsome clinical characterization of participants. Third, Studies 2 and 3 were not of a sufficient size to confirm the factor structure of the DARS. Yet, these studies provided convergent support for the internal consistency for this measure, as well as crucial support for concurrent, discriminant, and criterion validity. Fourth, Sample 3 included a self-report measure of depression severity. This may yield inflated estimates of the association between anhedonia and depression constructs due to their shared method variance and potentially incongruent ratings between self-report and clinician scales with higher depression severity (Corruble et al., 1999). Finally, for some of the analyses the sample size may have been too

small to yield definitive outcomes (e.g. differences across ethnicities). Future studies should confirm the correlation between DARS scores and depression based on either the Hamilton Rating Scale for Depression (Hamilton, 1960) or the Montgomery Asberg Depression Rating Scale (Montgomery and Asberg, 1979). Finally, test-retest reliability was not evaluated. This precludes conclusions about stability over time, or sensitivity to change.

In summary, the DARS is a promising scale to measure anhedonia with positive psychometric properties and has demonstrated incremental value over the SHAPS in evaluating interest, motivation and reward responsiveness as well as in defining subgroups, such as TRD. Future research should be aimed at confirming the factor structure of the DARS as well as cutoff scores for clinical use. In addition, the stability of the anhedonia construct within MDD should also be evaluated.

Conflict of interest

SJR, LQ, AC, RB and BAS have no financial disclosures to report. SHK has received grant/research support from: Bristol Myers Squibb, Canadian Institutes of Health Research, Clera Inc., Eli Lilly, GlaxoSmithKline, Janssen Ortho, Lundbeck, Ontario Brain Institute, Servier and St. Jude Medical. He is a consultant to AstraZeneca, BoehringerIngelheim, Bristol Myers Squibb, Eli Lilly, Lundbeck, Pfizer, Servier, and St. Jude Medical.

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Appendix

See Table A1 and A2.

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